

COURSE OUTLINE

ADDIS ABABA UNIVERSITY		College of Natural and Computational Sciences Department of Computer Science
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Course Title	Computer Graphics			
Course Code	CoSc3412			
Enrollment	Regular			
Program	B.Sc in Computer Science			
Credit hours	3	Lecture	Laboratory	Tutorial
		3	2	1
Credits	5 ECTS			
Course Category	Compulsory			
Module Code	CoSc-M4151			
Prerequisite(s)	CoSc2112 & CoSc3111			
Year	III			
Semester	II			
Teaching & Learning Methods	Lectures, Individual and group Assignments			
Assessment/Evaluation & Grading System	➤ Mid Exam- 25% ➤ Project Work -20% ➤ Assignment-10% ➤ Test-5% ➤ Final Exam- 40%			

Attendance Requirements	75%
Lecturer	Samuel G
Course Objectives & Competences	On completion of this course students should be able to: - Have a basic understanding of the core concepts of computer graphics. - Be capable of using OpenGL to create interactive computer graphics. - Understand a typical graphics pipeline. - Have made pictures with their computer. - have a good understanding of the role and importance of 2-D graphics in computer systems, - understand the mathematics underlying objects transformations and clipping operations, - be familiar to 3-D graphic concepts, - be familiar to surface rendering and aware of the techniques which can be used, - be able to produce photo-realistic rendering, - be able to draw simple objects using C/C++, and to model more complex objects using simple primitives, and - know how to create simple animations.
Course Description	Introduction (applications of computer graphics, Graphical User Interfaces, graphical input devices, graphical display devices, software standards, Open GL, GKS, PHIGS); simple drawing algorithms (line drawing and attributes, polygon drawing, circle drawing, ellipse drawing, character generation and attributes); transformations and clipping (basic transformations, translation, scaling, rotation, clipping algorithms); Windows and Viewports (co-ordinates representation, Viewing Pipeline, Window to Viewport transformation, zooming and panning effects with windows

	and viewports); 3-D Graphics (3-D Geometry, 3-D transformation, display methods, parallel projection, perspective projection, depth cueing, 3-D object representation, The Virtual Camera, modeling concepts); surface rendering.
Course Outline	Chapter 1: Introduction to Interactive Computer Graphics (4hr) 1.1. Brief History of Computer Graphics 1.2. 3D Graphics Techniques and Terminology 1.3. Common Uses of Computer Graphics 1.4. Application Area Chapter 2: Graphics Hardware (2hr) 2.1. Raster Display Systems 2.2. Introduction To The 3D Graphics Pipeline 2.3. The Z Buffer For Hidden Surface Removal Chapter 3: Introduction to The Rendering Process with OpenGL (3hr) 3.1. The Role of OpenGL in the Reference Model 3.2. Coordinate Systems 3.3. Viewing Using a Synthetic Camera 3.4. Output Primitives and Attributes Chapter 4: Geometry and Line Generation (5hr) 4.1. Point and Lines, Bresenham's Algorithm Generating Circles 4.2. Plotting General Curves 4.3. Line Thickness 4.4. Line Style 4.5. Polygons 4.6. Filling 4.7. Text and Characters Chapter 5: Geometrical Transformations (3hr) 5.1. 3D Transformation 5.2. Matrix Representation 5.3. Homogeneous Coordinates 5.4. Combination of Transformations Chapter 6: State Management and Drawing Geometric Objects (3hr) 6.1. Basic State management 6.2. Displaying Points Lines and Polygons

	6.3. Normal Vector 6.4. Vector Arrays Chapter 7: Representing 3D Objects (2hr) 7.1. Modeling Using Polygons 7.2. Techniques for Creating Representational Polygon Meshes 7.3. Non-Polygonal representations Chapter 8: Color and Images (2hr) 8.1. Color in Computer Graphics TGB, CIE 8.2. Image Formats and Their Applications: GIF, JPG, PNG Chapter 9: Viewing A local Illumination Model (4hr) 9.1. Using The Camera Model for Viewing 3D Scenes 9.2. Perspective and Other Types of Projection 9.3. Viewing Types of Light Source 9.4. Reflectance Models: Diffuse (Lambert) and Specular (Phong) Gouraud and Phong Interpolation 9.5. Lightning and Shading in OpenGL Textures Chapter 10: Application Modeling (4hr) 10.1. Distinction Between Modeling and Graphics 10.2. Immediate Mode Versus Retained Mode Model 10.3. Storage Strategies 10.4. The Matrix Stacks 10.5. OpenGL Display Lists: Traversal and Instancing 10.6. How The Concepts are Realized in Specific Systems: OpenGL and java3D
Text books & References:	Textbook - Donald Hearn, M. Pauline Baker: “Computer Graphics – C Version”, Pearson Education. References - Steven Harrington: “Computer Graphics – A Programming Approach”, McGraw Hill. - David F. Rogers: “Procedural Elements for Computer Graphics”, Tata McGraw Hill.

